

Network Analysis:



Edges vs. Social Ties

Social Ties are “Messy”

Edge vs. Social Tie

An edge in a graph is devoid of “meaning” or “content” Its very utility comes from context-free abstraction



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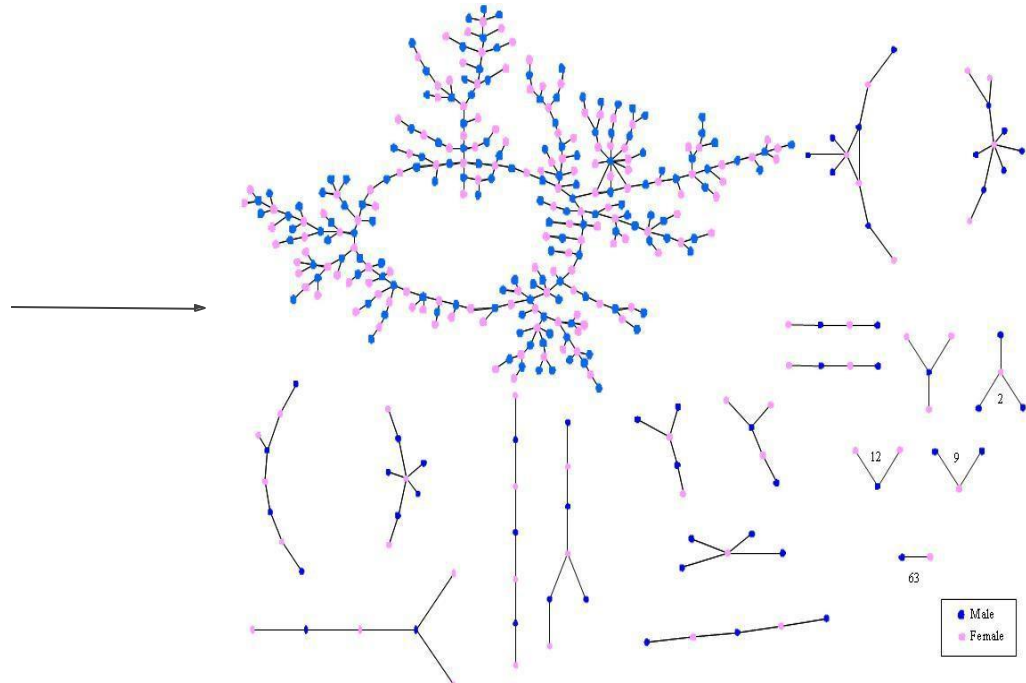
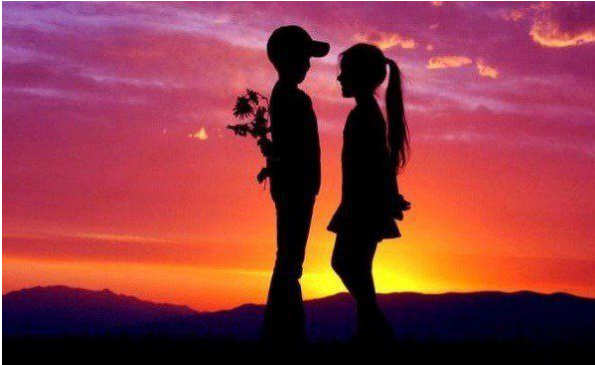


A social tie in a network is a hefty baggage: carries emotion, meaning, norms, expectations, trust, competition, social roles, and a history



Edge vs. Social Tie

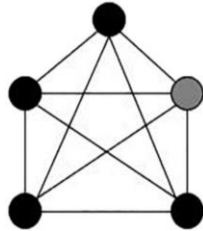
Yet, the messy, variegated content in that baggage can surprisingly constrain or enable the emergence of certain graph structures



Edge vs. Social Tie

The inverse is also true: a graph structure can also constrain or enable certain characteristics in social ties

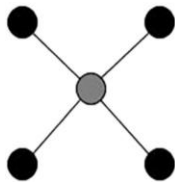
closed structure



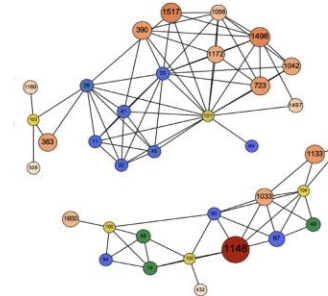
Monitoring, loyalty



open structure

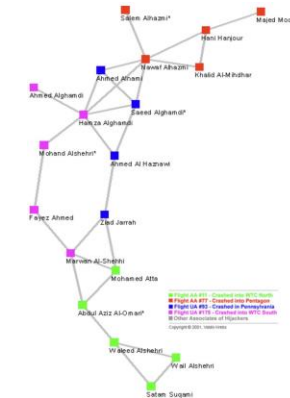


Easily fragmentable,
Low risk of annihilation



Mafia network

[Agreste et al. 2016](#)



Terrorist network

[Krebs 2002](#)

Social Ties and Social Support

People activate particular social ties for particular resources or support Strong vs. weak ties

- Strong ties generally provide wide range of support

Physical contacts (e.g., neighbors)

- Provide small/large services (e.g., borrowing sugar, giving a ride to the station)
- Limited emotional and financial support

Kinship ties (e.g., parents, siblings)

- Emotional and/or financial support

Abstraction: Social Ties as Edges

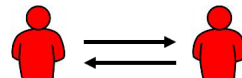
Reciprocity

The job of the network analyst: Apply or develop adequate abstract graph representations for a social tie. Example:

Reciprocity: The general social tendency to maintain **balance** in social exchange

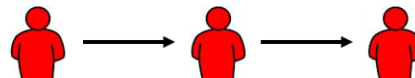
- Dyadic:

- i gives j resource x with value $v1$ at time $t0$
- j reciprocates with resource y with similar value $v2$ at $t1$



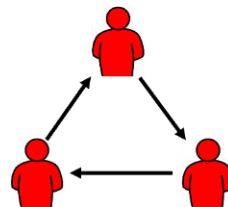
- Chain:

- Pay it forward (e.g., parental care)



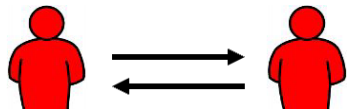
- Triadic:

- i gives to j (without expectation of direct reciprocation)
- j gives to k
- k gives to i



Measuring Reciprocity: Dyad Level

Examples:

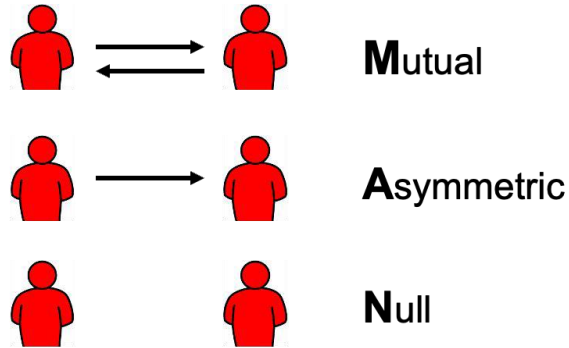


- How well balanced are the **values** of exchange between A and B?
 - $|v_1 - v_2|/2$
 - $|v_1 - v_2|/(v_1 + v_2)$
- How well balanced are the **times** to reciprocation?
 - $\Delta t_{i \rightarrow j} - \Delta t_{j \rightarrow i}$

Measuring Reciprocity: Network Level

Dyad Census

- Frequency of all dyadic isomorphism classes in a network
- A useful quantitative description of an observed network

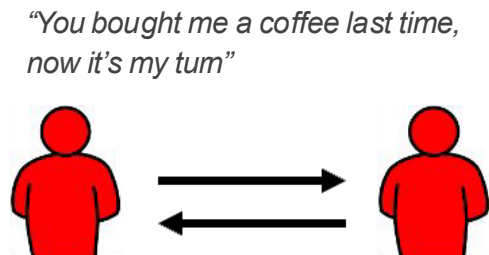


This approach makes it difficult to compare reciprocal tendencies between different networks

Measuring Reciprocity: Network Level

Index for Mutuality

- The **observed** numbers of M , A , N dyads are viewed as realizations of a **probabilistic** process governed by social forces
- Example: Societies with strong norms of reciprocity should have higher probability of mutual dyads
- Index of mutuality expresses such forces



Edges and Dyads

Index for Mutuality (ρ)

Core intuition: How much does the number of mutual dyads in the **observed network** (e.g., Japan's social network) deviate from the number of mutual dyads in a comparable **random network**?

Estimate a parameter ρ which quantifies the extent of this deviation In

statistical terms: The **probability of mutual dyads**:

$$P(i \rightarrow j \text{ and } j \rightarrow i)$$

Edges and Dyads

Index for Mutuality (ρ)

In statistical terms: We want to estimate the probability of mutual dyads:

Conditional probability: probability of i choosing j , multiplied by the conditional probability of j choosing i , when i chooses j


$$P(i \rightarrow j \text{ and } j \rightarrow i) = P(i \rightarrow j)P(j \rightarrow i | i \rightarrow j)$$



$$P(j \rightarrow i | i \rightarrow j) = P(j \rightarrow i) + \rho P(j \nrightarrow i)$$

$\rho=0 \rightarrow P(j \rightarrow i)$ and $P(i \rightarrow j)$ are independent (no reciprocation)

$\rho=1 \rightarrow P(j \rightarrow i | i \rightarrow j) = 1$, so every nomination is reciprocated

Social Ties and Diffusion

Social Ties and Information Diffusion

Similar to social support, people selectively talk about certain topics to certain types of relationships.

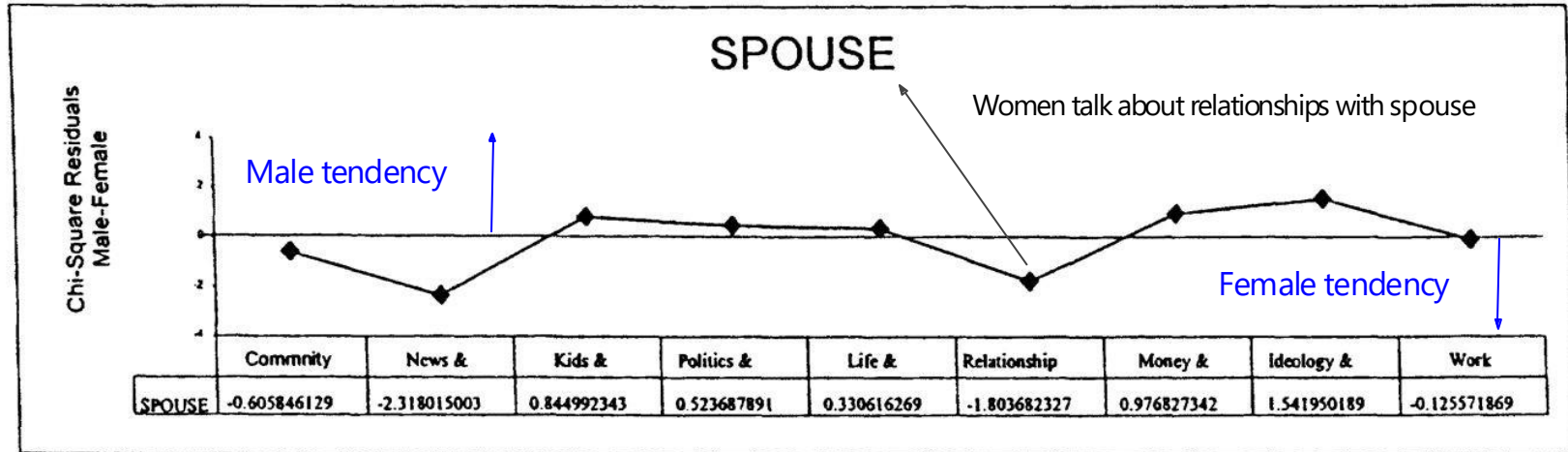
Examples:

- Sensitive topics (politics and religion) are usually discussed with close friends and family
- Generally, people discuss important matters with people they trust (i.e., confidants)
- Confidants potentially wield substantial influence on one's opinion
- At a more macro scale, studying opinion dynamics with confidant networks rather than an all-encompassing network might yield more insight

Social Ties and Information Diffusion

So, with **whom** do we discuss important matters? And **what** are those important matters?

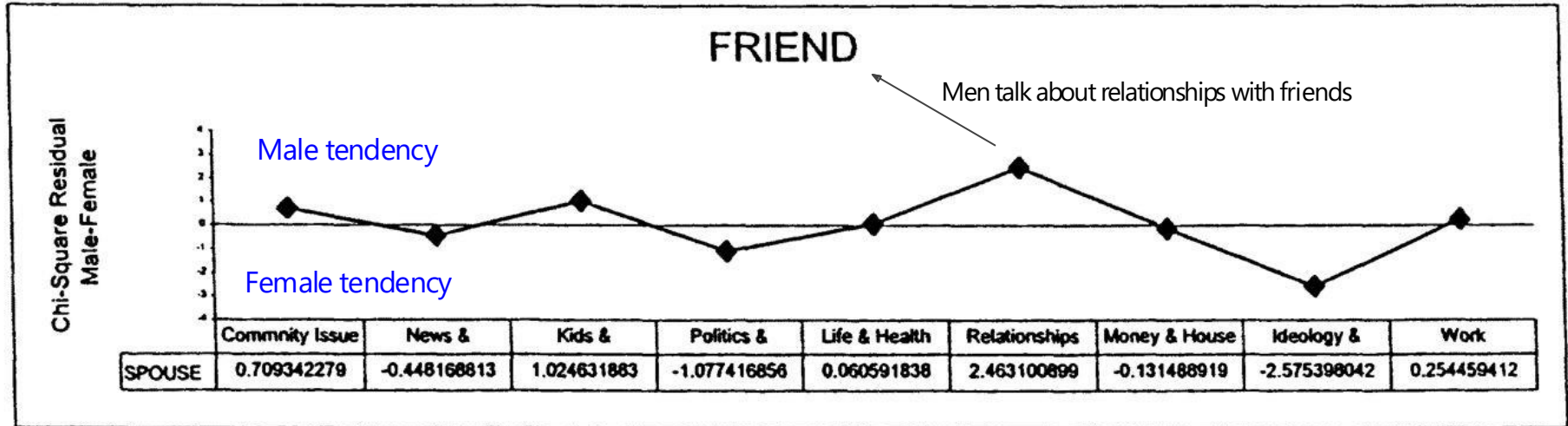
Panel a: Conversation Asymmetries for Talking with *Spouse*



Social Ties and Information Diffusion

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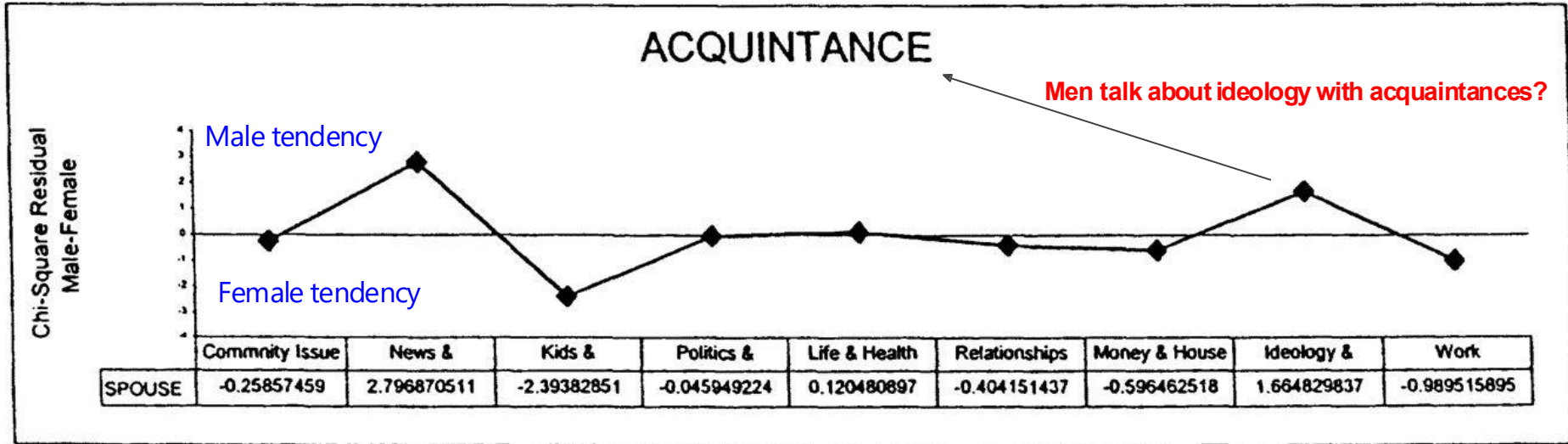
Panel b: Conversation Asymmetries for Talking with *Friend*



Social Ties and Information Diffusion

So, with **whom** do we discuss important matters? And **what** are those important matters?

Panel d: Conversation Asymmetries for Talking with *Acquaintance*



Social Ties and Information Diffusion

Men talk about ideology with acquaintances...

But, don't people discuss important, often private, topics with their trusted strong ties?

Answer: Not necessarily. "People may often confide in people they do not even consider confidants ([Small 2017](#))."

Social Ties and Information Diffusion

Why?

Strong ties (e.g., friends and family) are interconnected (i.e., triadic closure)

- Sensitive/embarrassing information disclosed to a friend can quickly spread to other close friends in the same social circle



Social Ties and Information Diffusion

Why?

Strong ties (e.g., friends and family) are interconnected (i.e., triadic closure)

- Sensitive/embarrassing information disclosed to a friend can quickly spread to other close friends in the same social circle

Topic-alter dependency

- Strangers share very few social contexts, so people feel safe to disclose sensitive topics



Social Ties and Information Diffusion

So, will word about the exquisite cake from La Gourmandine spread like wildfire at the party?

Not necessarily, because **valued information** about scarce goods might spread through highly exclusive **strong ties**

Alternatively, gourmet cake as a topic might be discussed between cake lovers, but not with, say, people who are sensitive to gluten/dairy or who are indifferent → **topic-alter dependency**



Social Ties and Information Diffusion

Network science provides powerful tools for modeling information diffusion.

Yet, if the ties are inadequate for the phenomenon under study, network analysis will be irrelevant.

Hence, qualitative aspects of social ties (the “messy” content) must be carefully evaluated:

- Types of social ties for constructing the network
- Strength of ties
- Topic-alter dependency

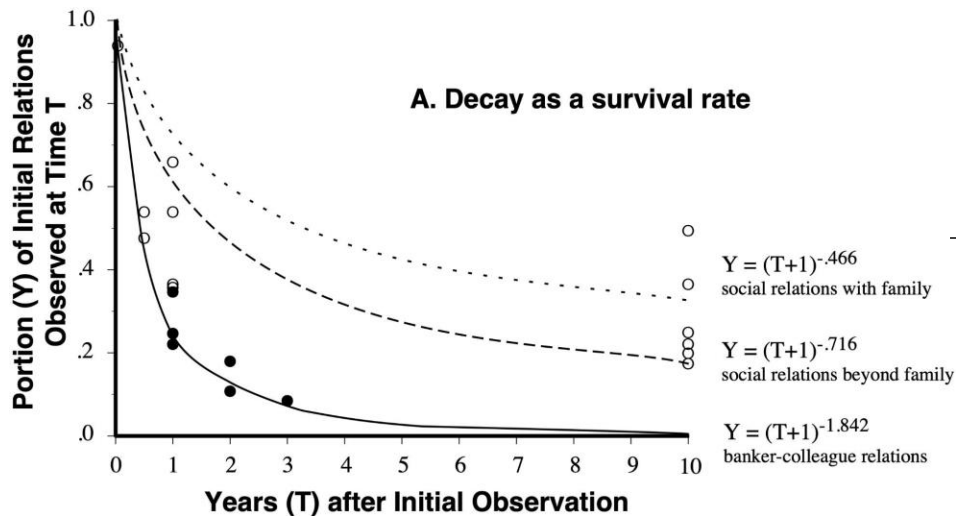
Edges

The Dynamics of Social Ties

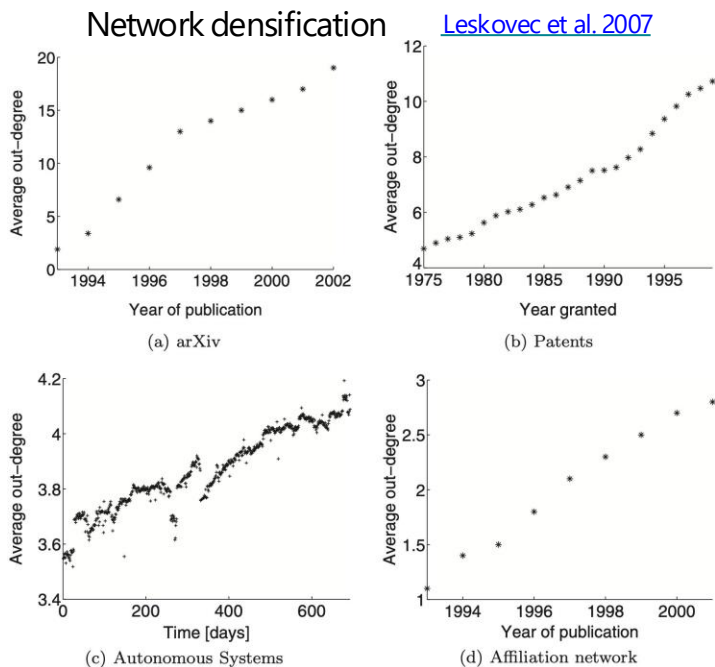
Persistence and Decay of Social Ties

People form relationships and those relationships can persist or subside over time

The evolution of a social network is closely related to such ebbs and flows of social ties



[Burt 2000](#)



Interdependence and Persistent Social Ties

Then what factors influence how long a tie persists (commitment to a relationship)?

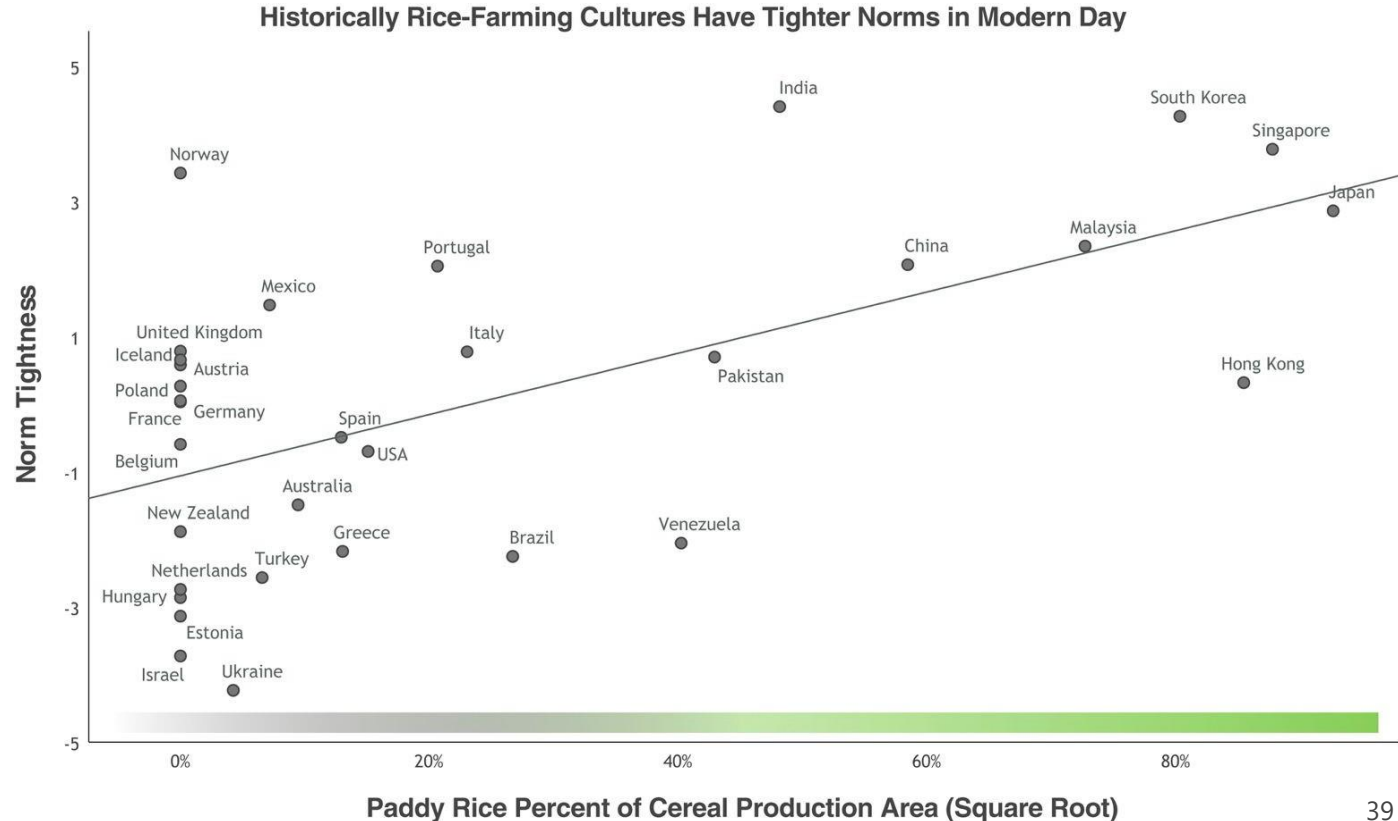
- Historically, more interdependent modes of production seems to have influenced people's thinking styles and social organization, including how people relate with one another.

Rice farming requires highly interdependent, coordinated labor, compared to wheat farming.



Interdependence and Persistent Social Ties

Even across countries, rice farming cultures have “tighter” norms – stronger group pressure on individual conformity.



Uncertain Environments and Social Ties

These historical differences may have contributed to systematic differences in generalized trust and commitment to relationships

Survey of Japanese and American respondents

Q: "Do you think you can put your trust in most people, or do you think it's always best to be on your guard?"

A: "People can be trusted" **47% American vs. 26% Japanese**

Uncertain Environments and Social Ties

Japanese society enforces stricter norms within groups, which provide security to their members.

- Strong trust for in-group members (norm violation is met with harsh sanctions)
- Much weaker trust to outsiders/strangers (relatively weaker norms to ensure security)

In the extreme, if everyone distrusts outsiders, individually optimal choice is to rather stay in the community and increase commitment to existing ties.

- Strong ingroup trust: low transaction cost
- Static relationships: high opportunity cost



Uncertain Environments and Social Ties

Individualist cultures (e.g., U.S.) where the environment forced self-sufficiency and lower interdependent modes of subsistence (think the wild west):

- Necessary to learn to trust strangers
- High transaction cost (due to thin trust)
- Low opportunity cost (possibility of discovering more beneficial interactions)



Uncertain Environments and Social Ties

It is not so much a matter of culture:

It is more a matter of **structure**

- In a society where **in-group cohesion** is strong, general trust becomes less critical

It is also more a matter of circumstances

- Does the environment force interdependent modes of subsistence?
- Is there high uncertainty in the environment?

Summary

An interpersonal tie influences and is influenced by the broader network structure

- Social support differs by type of relationship
- Topic-alter dependency can affect information diffusion
- Social tie can create a graph signature
- Reciprocity
- Dynamics of social ties hold implications for network structure